

P44 - Evidence-based provenance scoring for creative digital media content

Group Formal Tag: COMP3888_M17_01_ZL

Tutor Name: Zhenlin Liu

Client Name: Alt.VFX

Group members details

(indicate if any member is doing multiple units)

No.	Student Name	Student ID	Student Unikey	XP-Roles
1	Declan Schulz	540894767	dsch0965	
2	Hongjie Deng	540227655	hden0671	
3	Khanh Phan	530800451	dpha0698	
4	Marsya Amanda	530344829	mama0048	
5	Nick Chen	540658848	nche0916	
6	Shuhao Zhang	540740693	szha0571	
7	William Alexander	540722172	wale0194	
8	Yee Cheng Ng	540444920	yeng0449	

Attendances: *Declan Schulz, Hongjie Deng, Khanh Phan, Marsya Amanda, Nick Chen, Shuhao Zhang, William Alexander, Yee Cheng Ng*

Apologies: N/A

Submission Date: 30/08/2026

UG Capstone Project Weekly Meeting Minutes

Weekly Group Meeting Minutes

Time: 26/08/26

Venue: Zoom

Meeting Minute Taker: William Alexander

#	Agenda Item	Description/ Comments	Decision/Action	Who?	Items for escalation
1	What has been completed?	Discussed using available audio datasets as test data.	Prioritise further research available data and its potential utility for scoring system	All members	Starting initial code
2	What is in progress?	Creation of evidence evaluation schema	Start by creating table representing evidence strength & utility	All members	Collaborate with other teams for input/output schema standardisation
3	What is working well?	Communication has been stronger than last week	Continue to strengthen communication.	All members	Plan roles as development begins.
4	What needs improvement?	Communication regarding implementation details	Delegate tasks and add summaries & notes on shared documents	All members	Establish a leader for implementation plans
5	Reminders	We will have an additional meeting before the client meeting on Thursday this week	Sent follow up in group chat	All members	N/A
6	What needs to be completed before the next meeting?	Complete research and checking on client-provided datasets, review	Everyone sets time to research independently	All members	How to determine links between

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		other team's proposed schema			provided evidence
7	What's Next	Complete sparkPLUS peer review	Collaborate with CSEC security team.	All members	Collaborate with other teams to begin shared schema programming
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Main Discussions

- Research evidence classes (each person research the evidence class assigned)
 - Metadata
 - Audio content (?) - TBD, dependent on research
 - Output: produce a SHORT description, data map, and human vs. AI audio feature of the evidence class (if you can find any)
- Research audio mixing/mastering/recording process - Mars
 - Research how AI is being used in the creative process (not final output)
 - Look at client resources on how media studios use AI for audio prod
 - Output: gather and summarise 3-5 high-quality sources which explain the weight approximations for each evidence class, and report any correlated evidence classes
- Reach out to CSEC to produce adversarial dataset - Declan
- Focus on reverse-engineered data or AI generated music?
- Decide schema w other group (Hong)

Key Takeaways

- Need a better idea of the evidence classes we're dealing with
- Need to have a plan of how to represent evidence
 - Should finalise inputs/outputs for CSEC team testing

What Happened

- Concluded that more work needs to be done regarding implementation plan and details
- Delegated dataset research to different team members

What's Next

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- ☐ Assigning who during the next week will take meeting minutes
- ☐ Identifying what deliverables are to complete in the upcoming week and assign work
- ☐ Organising an additional meeting for the upcoming project presentation

Weekly Client Meeting Minutes

Time: 27/08/26

Venue: Zoom

Meeting Minute Taker: William Alexander

Attendances: Declan Schulz, Hongjie Deng, Khanh Phan, Marsya Amanda, Nick Chen, Shuhao Zhang, William Alexander

Apologies: Yee Cheng Ng

#	Agenda Item	Description/ Comments	Decision/Action	Who?	Items for escalation
1	What has been completed?	Updated client with our latest progress and got client's feedback and ideas.	Client confirmed the open-ended nature of the scoring system – This it up to our discretion and must be justifiable	All members	
2	What is in progress?	Collaborating with other teams for shared schema, developing scoring methodology	Wait for github creation for common schema	All members	Once schema finalised, communicate with CSEC group about testing
3	What is working well?	We had 2 meetings preceding the client meeting to ensure we were	Continue regular meetings before client meeting	All members	Ensure meaningful progress is done

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		well-prepared for the meeting.			before client meeting.
4	What needs improvement?	Communication with other teams working on the project	Collaborate more on Discord and engage in collaborative development on Github for shares schema	All members	Ensure group is familiar with proposed schema and check Discord frequently so development can start as soon as repo is released.
5	Reminders	Next weeks meeting will resume during regular time as discussed	Set reminder in group chat of meeting time	All members	N/A
6	What needs to be completed before the next meeting?	Start development of shared schema and further develop evidence scoring.	Internally prioritise development of these.	All members	Delegate roles for evidence scoring development.
7	What's Next	Think of any other questions that we might need to clarify client wants/needs	Prepare beforehand any clarifications we want and also any topics to discuss between other capstone groups.	All members	N/A
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Main Discussions

- Updated client with latest progress:
 - o Initial development of evidence scoring schema
- Discussed shared schema with other groups

Key Takeaways

- Score vector: completely up to team's discretion to create & justify
- Development of shared schema should start soon, in the next few weeks

What Happened

- Shared team progress
 - o Discussed shared schema development
- Watched other team's progress updates and demos

What's Next

- Engage with other teams to develop shared schema
- Reach out to CSEC team for collaboration

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Weekly Tutor Meeting Minutes

#	Agenda Item	Description/Comments	Decision/action	Who	Items for escalation
1	What has been completed	Presentation of project proposal	Presented project proposal; Got feedback from tutor regarding project	All members	Ensure any tutor concerns with the project are addressed
2	What is in Progress	Completion of sparkPLUS evaluation	Complete sparkPLUS evaluation	All members	Ensure evaluation is complete by deadline
3	What is working Well?	Communication between team and tutor. Tutor is happy with our progress with the project	Continue as we are with communication channels	All members	Keep meeting weekly with tutor
4	What needs improvement?	N/A	N/A	All members	None
5	Reminders	N/A	N/A	All members	None
6	What needs to be completed before next meeting?	sparkPLUS evaluation	Everyone should complete their individual evaluation by Monday	All members	Everyone should complete their individual evaluation by Monday
7	What's next?	sparkPLUS evaluation	Set reminder in calendar that everyone needs to complete sparkplus	All members	Remind everyone to complete sparkplus

Main Discussions

- Sparkplus rating system needs to be 4 across all team members
 - o This means everyone did equal amount of work
 - o Do not put all 7 or 1 as this will invalidate your rating
- Tutor is happy with our progress in the project
 - o Confirmed that the project is software development rather than research

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Key Takeaways

- Confident in the trajectory of our project

What Happened

- Discussed upcoming due dates and how to fill in sparkplus
- Discussed our performance in the presentation

What's Next

- Continue making effort to meet with tutor and ensure progress in the project

Individual Progress Report

Name: Declan Schulz

Student ID: 540894767

Goals

- Research DAW session files
- Look through the Linux MultiMedia Studio Sharing Platform dataset to see what data can be parsed
- Make a judgement on the usefulness of DAW session files as an evidence source

Progress Made

- Read about DDEX and other industry standards for session credits (useful information, but incredibly easy to forge)
- Read about DAW session files, what metadata they store and how accessible it is (varies greatly from DAW to DAW, but is often hard to access and easy to forge)
- Downloaded and looked through the LMSS Platform dataset

Blockers to Progress/Issues

- DAW session files are a bit of a dead end, not sure where to look next
- Since there's such a vast amount of evidence types and links between evidence, it's hard to narrow down our scope

Individual Progress Report

Name: Hongjie Deng

Student ID: 540227655

Goals

1. Parse the provided dataset and collect the variables that are needed in the evidence schema.
2. Discuss with other teams about the general evidence schema for every domain.
3. Build a parser for wav file

Progress Made

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1. I am successful to extract the metadata from one of the given datasets, and pinned down some key variables
2. I am successful to build a .wav file parser

Blockers to Progress/Issues

1. We are still discussing with other teams about the general evidence schema, as we want to create an evidence schema with sufficient variables to support the evidence chain when evaluating the evidence relationship

Individual Progress Report

Name: Khanh Phan

Student ID: 530800451

Goals

- Look into the client given data set such as stem and comp
- Find datasets with DAW session files
- Extract the file info such as fps, bytes, second for future comparison
- Plan out how we can use the data human work analysis

Progress Made

- I looked through all the dataset and confirmed what file we have to deal with
- I build a prototype for extracting file data
- I got a rough idea on how I can score the dataset

Blockers to Progress/Issues

- Scheduling issues, communication delays

Individual Progress Report

Name: Marsya Amanda

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Student ID: 530344829

Goals

- Research AI-assisted (and otherwise) music mixing/production workflows and discuss research findings with client
- Explore dataset provided by client and investigate the relevant evidence class (stems)
- Note how stems may be affected by AI and how to map to final artefact
- Collaborate with other groups on making a unified schema
- Find datasets with DAW session files

Progress Made

- I collected and summarised findings on music production workflows
- Found dataset on DAW session file, not sure if I'd use it
- Found characteristic of stems in regards to AI modification
- Started to collaborate with other team on unified schema but no concrete agreements yet

Blockers to Progress/Issues

- Scheduling issues, communication delays
- Disagreements (internally and with other groups) on how to implement the project

Individual Progress Report

Name: Nick Chen

Student ID: 540658848

Goals

- Decide on what components should go into the file and output schema
- Research what components we want to contribute to our score vector, ie what features to score against
 - Look into existing schemas and see whether we can build on them
- Think of how to consider implicit attestation element, ie how to take into account what the creator is actually attesting about their submitted evidence

Progress Made

- Need to decide on a schema and communicate with cyber security team how to integrate their workload with ours
- Work with other groups to finalise input and output schemas for cross compatibility
 - o Confirming whether they want to use a shared library

Blockers to Progress/Issues

- Fixing client meeting time, this has changed as they have been overseas
- Needing to keep up/pay more attention to communication across groups to check on how they want to approach/develop the schema

Individual Progress Report

Name: Shuhao Zhang

Student ID: 540740693

Goals

- study about Voice memos and demos as my evidence class for the project
- Conduct initial analysis of the TELEFUNKEN multitrack dataset.
- update to tutor about our next week plan

Progress Made

- Conducted initial analysis of the TELEFUNKEN dataset and identified potential audio evidence relationships.
- discuss with teammate about what we will discuss next week and update to our tutor

Blockers to Progress/Issues

- hard to get update from other student and our work looks bit too separate

Individual Progress Report

Name: William Alexander

Student ID: 540722172

Goals

- Research the Cambridge-MT Mixing secrets dataset
- research and test methods to match audio waveforms

Progress Made

- Downloaded examples from Cambridge-MT mixing secrets dataset
- created demo for basic sound-matching algorithm

Blockers to Progress/Issues

- Many false-positives for sound matching algorithm

Individual Progress Report

Name: Yee Cheng Ng

Student ID: 540444920

Goals

- Visualise the overall project architecture and dataset breakdown.
- Clarify project expectations and requirements.
- Research approaches for parsing and standardising different evidence formats.
- Conduct initial analysis of the TELEFUNKEN multitrack dataset.

Progress Made

- Conducted initial analysis of the TELEFUNKEN dataset and identified potential audio evidence relationships.
- Implemented and evaluated RMS-based analysis for cross-evidence consistency between audio tracks.
- Researched potential machine learning and audio analysis techniques that could support further evidence validation.

Blockers to Progress/Issues

- Coordination and communication with other collaborating teams working on related parts of the project.
- Resolving dependencies between different components and tasks within the overall system architecture.